Section	Subsection	Paragraph role and one-sentence summary
Abstract	–	• Whole-review synthesis — Tile drainage is presented as a fundamental hydrological intervention that reshapes flow and contaminant export, creates substantial but manageable nitrate and phosphorus losses, supports effective multi-level mitigation, provides major agronomic benefits, and can now be understood and managed using mature modeling and mapping tools.
Introduction	–	• Establish scope and importance — Tile drainage is geographically extensive and agriculturally important, and by rapidly rerouting shallow groundwater and runoff to streams it functions as a master control on the hydrology and water quality of intensively farmed landscapes. • Frame the central trade-off — Drainage reduces waterlogging, surface runoff, sediment, and some phosphorus losses while simultaneously opening rapid pathways for nitrate and other contaminants, creating a fundamental trade-off between agricultural productivity and downstream water quality. • Establish the evidence base — The review draws on complementary monitoring, tracer, experimental, long-term, modeling, geophysical, and remote-sensing evidence, with confidence strongest where these independent approaches converge. • Define scope and organization — The paper focuses on established knowledge and follows a causal sequence from hydrology through contaminant transport and management to agronomic outcomes and methodological tools, while unresolved questions are assigned to a companion frontier review.
Drainage Hydrology: Flow Pathways, Regime, and Scaling	–	• Section framing — Subsurface drainage fundamentally restructures field and catchment water balances by rapidly routing water to streams, producing characteristic flow regimes and cross-scale hydrological behavior that underpin the subsequent water-quality and management effects.
	Tile Drainage as a Dominant Flow Pathway	• Quantify tile-flow dominance — Field, watershed, tracer, and modeling studies consistently show that tile drains carry a large fraction of precipitation-derived water and commonly account for roughly 40–65% of streamflow in humid drained landscapes. • Show event and scale dependence — The fraction of discharge routed through tiles varies systematically between event and non-event conditions and changes with drainage area, demonstrating that tile–surface partitioning is scale dependent. • Establish seasonal controls — Tile contributions are greatest when soils are wet, evapotranspiration is low, snowmelt or seasonal recharge occurs, and the water table rises sufficiently relative to drain depth.
	Modification of the Flow Regime	• Establish low-flow and baseflow effects — A robust cross-study finding is that drainage increases low flows and baseflow contributions by converting slowly stored subsurface water into sustained stream discharge. • Characterize peak-flow response — Drainage has relatively modest effects on the largest floods but can either increase or decrease smaller and intermediate peaks depending on antecedent conditions, spatial scale, drainage placement, and event magnitude.

		• Qualify flashiness and drought effects — Effects on flashiness and drought behavior depend on soil, slope, drainage intensity, and temporal resolution, although drainage generally alters low- and medium-flow behavior more consistently than extreme flows.
	Scale Dependence, Homogenization, and Travel Times	• Establish hydrologic homogenization — Artificial drainage reduces the natural scale dependence and spatial variability of catchment response, making heavily drained basins behave more uniformly while shortening and stabilizing travel times. • Quantify travel-time shortening — Modeling, analytical solutions, and tracer studies demonstrate that increasing drainage density or incision substantially shortens groundwater travel and riparian residence times while increasing the fraction of groundwater intercepted by tiles. • Identify pathway-activation thresholds — Tile and surface runoff are activated by combinations of rainfall, antecedent soil moisture, water-table position, and preferential connectivity rather than by rainfall magnitude alone. • Partition fast and slow flow — Tracer and geochemical evidence shows that tile discharge combines rapid preferential flow with slower matrix and pre-event water, with their relative contributions varying by wetness, season, and soil texture.
	Repartitioning of Water Relative to Undrained Land	• Compare drained and undrained partitioning — Compared with undrained or surface-drained land, improved subsurface drainage generally reduces surface runoff, sediment, and phosphorus while increasing subsurface water export and nitrate delivery. • Establish the dominant field outlet — At field scale the drain is usually the dominant water-export pathway in humid landscapes, although the magnitude and associated phosphorus consequences depend strongly on soil texture and infiltration behavior.
Nitrate Export: Magnitudes and Biophysical Controls	–	• Section framing — Nitrate is the dominant nitrogen form exported from tile-drained cropland, with losses primarily routed through drains and controlled by hydrology, rapid preferential transport, nitrate sources, and incomplete denitrification.
	Subsurface drainage as the dominant nitrogen pathway	• Establish the dominant N pathway — Evidence from field to watershed scales consistently demonstrates that subsurface drainage carries most nitrogen export and that nitrate constitutes the dominant fraction of that exported nitrogen. • Quantify loads and concentrations — Drainage nitrate concentrations and annual loads are frequently high and often exceed drinking-water standards, while interannual load variability is driven much more strongly by water export than by concentration. • Identify source and legacy controls — Regional nitrate export reflects interacting fertilizer inputs, hydrologic throughput, mineralization–uptake asynchrony, and persistent soil-N pools that can delay responses to management for years or decades.
	Hydrologic control of nitrate export and seasonality	• Establish discharge control — Across scales and drainage designs, annual nitrate loss is strongly related to drainage discharge, making water yield the dominant explanation for year-to-year variation.

		• Establish event and seasonal dominance — A disproportionate share of nitrate is exported during high-flow events and the nongrowing season, allowing short hydrological episodes to account for much of the annual load.
	Concentration–discharge behaviour, sources, and denitrification	• Diagnose transport using C–Q behavior — Chemostatic concentration–discharge behavior, counterclockwise hysteresis, rapid tile responses, preferential flow, and moisture thresholds indicate strong and rapid hydrological connectivity between nitrate source zones and drains. • Identify nitrate sources and attenuation — Isotope and geochemical evidence indicates that much drainage nitrate originates from nitrification of organic N sources, while denitrification can substantially attenuate nitrate where suitable flow paths occur but is often bypassed by drainage networks.
Phosphorus and Sediment Loss and Preferential Transport	Overview	• Reframe subsurface P transport — The traditional view of the soil profile as a clean subsurface filter is replaced by evidence that tiles can dominate field phosphorus loads through rapid preferential transport, with soil P status, legacy P, macropores, and episodic hydrology controlling losses.
	Phosphorus Loss Pathways and Water Partitioning	• Contrast concentration and load pathways — Although surface runoff often has higher phosphorus concentrations, the much larger water volume carried by tiles means subsurface drainage frequently dominates annual dissolved reactive phosphorus and total phosphorus loads. • Define exceptions to tile dominance — Tile dominance is not universal, particularly in frozen, snowmelt-driven, clay-rich, pasture, or flooded systems where surface runoff can remain the primary phosphorus pathway.
	Particulate versus Dissolved Phosphorus and Bioavailability	• Establish P speciation — Particulate phosphorus frequently constitutes the majority of tile-drain total phosphorus, although dissolved forms dominate in particular snowmelt, grassland, organic-soil, and high-soil-P environments. • Emphasize bioavailable P — Because eutrophication depends on biologically available phosphorus, both dissolved P and desorbable particulate P must be considered rather than relying on total P or a single analytical fraction.
	Soil Controls on Phosphorus Release and Legacy Phosphorus	• Identify soil-P and sorption controls — Dissolved P loss increases strongly with soil-test P and degree of P saturation, especially where high-P topsoil overlies subsoil with weak sorption capacity. • Establish legacy-P persistence — Long-term increases in soluble reactive phosphorus despite declining contemporary P balances demonstrate that accumulated legacy P and enhanced hydrological connectivity can sustain watershed export. • Identify high-risk organic soils — Drained organic and peat soils can become exceptionally large phosphorus sources because wetting–drying, mineralization, and redox processes mobilize stored P.
	Preferential Flow, Macroporosity, and Drainage-Induced Soil Structure	• Identify biopores as rapid conduits — A very small volume of continuous earthworm burrows, cracks, root channels, and installation-related macropores can dominate rapid movement of water and solutes from the surface to drains. • Describe structural gradients near drains — Macroporosity generally decreases with depth and can be locally enhanced near

		drain lines, while drainage modifies bulk density, aeration, diffusivity, and biological activity in context-dependent ways. • Explain long-term structural evolution — Drainage, loosening, and mole systems progressively increase hydraulic conductivity and can drive substantial pedogenic and redox-driven changes in soil structure over years to decades. • Demonstrate rapid macropore transport — Tracer and manure studies show extremely rapid but connectivity-limited transport to tiles, and realistic simulation therefore requires dual-domain or explicit macropore–drain representations.
	Temporal Dynamics of Phosphorus and Sediment Loss	• Establish seasonal and event dominance — Most phosphorus loss occurs during nongrowing-season and high-flow events, with concentration–discharge behavior indicating an increasing particulate contribution to tile export at high discharge. • Trace particulate sources and sampling bias — Tracer and fingerprinting studies show that much tile-exported particulate matter originates rapidly from topsoil, making episodic events disproportionately important and causing low-frequency sampling to underestimate annual losses.
	In-Field Controls on Phosphorus Loss	• Synthesize in-field P trade-offs — Long-term P inputs and surface applications increase drainage loss, while tillage choices often trade dissolved against particulate P, so effective management must match practices to site hydrology, connectivity, weather, and legacy P.
In-Field Nutrient and Cropping Management	–	• Section framing — In-field nitrate management acts primarily through nitrogen rate, source and timing, crop choice, seasonal drainage, and soil N processing, with multiple complementary practices generally needed to reduce losses without sacrificing production.
	Nitrogen Rate, Source, and Residual Soil Nitrate	• Establish the N-rate response — Nitrate loss increases consistently with fertilizer rate while crop yield approaches a plateau, making application near the economic optimum a central strategy for avoiding disproportionate environmental losses. • Explain residual-N legacies — Excess fertilizer creates residual soil nitrate that can accumulate during dry periods and be flushed during later wet years, generating multi-year nitrogen legacies. • Compare fertilizer sources — Manure often produces greater drainage nitrate than comparable inorganic fertilizer, especially with repeated or fall applications, although the magnitude varies among studies.
	Nitrogen Timing, Placement, and Enhanced-Efficiency Products	• Evaluate application timing — Moving nitrogen from fall to spring or in-season application generally reduces nitrate loss while maintaining yield, although weather and drainage volume can outweigh the timing effect. • Evaluate placement and efficiency products — Split application, placement, nitrification inhibitors, and other enhanced-efficiency approaches show inconsistent effects, whereas polymer-coated products and reduced-rate targeted applications provide more reproducible multi-pathway benefits in some settings.
		• Establish cover-crop effectiveness — Winter cereal rye is one of the most consistently effective in-field practices, commonly lowering nitrate concentration substantially and

		producing large load reductions when establishment is successful.
	Cover Crops: Reductions, Mechanisms, and Yield	• Explain the cover-crop mechanism — Cover crops reduce nitrate mainly by assimilating mineral nitrogen during the fallow period, with effectiveness strongly dependent on biomass, establishment timing, climate, and weather rather than major reductions in annual drainage volume. • Evaluate yield and overall performance — Well-managed rye generally produces little consistent yield penalty while clustering around roughly 40–60% nitrate-load reduction, although maize, drought, and poorly established systems present important exceptions.
	Crop Rotation, Diversification, and Perennial Systems	• Compare crop rotations — Cropping system strongly affects drainage nitrate, with diversified rotations generally producing lower losses and better yield-scaled performance than continuous corn. • Establish the perennial-system benchmark — Prairie, alfalfa, CRP, and other perennial systems produce extremely low nitrate concentrations and loads compared with annual row crops, although benefits can disappear rapidly after reconversion.
	Tillage and Residue Management	• Assess tillage and residue influence — Tillage and residue management exert relatively small and inconsistent effects on drainage nitrate compared with precipitation, crop rotation, and nitrogen rate.
	Soil Testing, Use Efficiency, Regional Aggregation, and Loss-Pathway Tradeoffs	• Scale soil-testing programs — Soil-test-based fertilizer programs can substantially reduce nitrate concentrations and residual nitrogen while maintaining yield, although large watershed reductions still require broader cropping-system changes and adoption support. • Estimate regional mitigation potential — Scaling field results across the Corn Belt indicates that widespread or spatially targeted rye adoption could produce substantial regional nitrate reductions at measurable but potentially competitive costs. • Identify pollution swapping and stacking — Because nitrate practices can redistribute nitrogen among leaching, N_2O, and NH_3 pathways, multi-pathway evaluation and combinations of rate optimization, timing, cover crops, diversified rotations, and drainage management are preferable to reliance on any single intervention.
Drainage Design and Water-Table Management	–	• Section framing — Drain geometry and water-table operation provide direct control over drainage volume and therefore nutrient loads, linking physical system design to environmental performance.
	Drain Spacing and Depth as Controls on Flow and Loss	• Quantify spacing and depth effects — Closer and deeper drains generally increase drainage volume and nitrate load without consistently changing concentration, while reduced drainage intensity can shift water and phosphorus toward surface runoff. • Derive the optimum-intensity principle — The evidence supports using only the drainage intensity needed to relieve crop waterlogging, because shallower or wider systems can substantially reduce nitrate loss at relatively small production costs.

	Controlled Drainage and Drainage Water Management	• Establish controlled-drainage effectiveness — Controlled drainage consistently reduces subsurface outflow and nutrient losses across climates, often with modest yield benefits, although its magnitude depends on site and operating strategy. • Quantify field-scale performance — Long-term field studies commonly report substantial reductions in drainage volume and nitrate export under controlled systems, frequently accompanied by maintained or increased yields. • Explain the flow–load mechanism and trade-offs — Controlled drainage reduces nitrate primarily by reducing discharge rather than concentration or greatly enhancing denitrification, with potential surface-phosphorus trade-offs and strong sensitivity to control timing.
	Drainage Water Recycling and Subirrigation	• Establish recycling co-benefits — Drainage water recycling simultaneously reduces downstream nutrient export and provides stored water that stabilizes or increases crop yield during drought and excess-moisture years. • Define subirrigation boundary conditions — Subirrigation can substantially reduce nitrate and improve yields by maintaining root-zone water, but its effectiveness is strongly constrained by soil hydraulic conductivity and local hydrogeology.
	Enhanced Configurations and Persistence under Climate Change	• Survey enhanced drainage configurations — Multi-level drainage, permeable filters, combined ditch–pipe systems, siphons, artificial macropores, and supplementary drainage extend conventional design principles by modifying drawdown, runoff, salinity, and hydraulic performance. • Assess climate robustness — Modeling across multiple regions indicates that controlled drainage retains substantial effectiveness under projected warmer and wetter climates and may become increasingly valuable during future growing seasons.
Edge-of-Field Treatment, Reactive Media, and Watershed-Scale Integration	–	• Section framing — Edge-of-field technologies intercept nutrients that have already entered drainage networks, with treatment performance controlled primarily by residence time, temperature, flow, loading, chemistry, and the proportion of drainage actually captured.
	Constructed wetlands	• Establish wetland treatment performance — Constructed wetlands reliably remove substantial nitrate through denitrification when appropriately sized and hydraulically loaded, although phosphorus retention is much less consistent.
	Woodchip bioreactors: performance, controls, and mechanisms	• Quantify bioreactor field effectiveness — Full-scale woodchip bioreactors consistently remove nitrate across climates, but annual edge-of-field reductions vary widely because removal efficiency and the fraction of flow treated differ among installations. • Identify hydraulic and temperature controls — Bioreactor efficiency increases with hydraulic residence time and temperature but trades off against flow capture, while increasing nitrate loading tends to increase removal rate but reduce percentage removal. • Explain the denitrification mechanism — Heterotrophic denitrification is overwhelmingly the dominant nitrate-removal mechanism, sustained by wood-derived carbon and microbial

		communities whose composition varies with substrate, temperature, depth, and season. • Translate controls into design — Bioreactor sizing must balance residence time against annual flow capture, with long narrow beds and appropriate cross-sectional area generally improving hydraulic efficiency and a variety of enhancements extending treatment capacity.
	Pollution swapping and greenhouse-gas by-products of bioreactors	• Assess treatment by-products — Bioreactors can generate N2O, CH4, sulfate reduction, and dissolved organic carbon, but N2O emissions are generally very small relative to nitrate removed and most undesirable by-products occur under excessive residence time or startup conditions.
	Reactive media for phosphorus removal	• Introduce P-specific reactive treatment — Because woodchips remove little phosphorus, reactive filters target dissolved phosphorus primarily through iron-based adsorption or calcium-based precipitation. • Compare reactive-media mechanisms — Iron-based materials generally provide the highest and most stable phosphorus removal over varying pH, whereas calcium-rich materials can perform strongly until reactive calcium and alkalinity are depleted. • Assess field performance and longevity — Phosphate uptake can be rapid and some iron media are regenerable, but field performance depends strongly on hydraulic capture and longevity, making combinations of phosphorus media and woodchip treatment attractive for simultaneous nitrogen and phosphorus control.
	Saturated buffers, denitrification walls, ditches, and in-channel practices	• Evaluate alternative interception practices — Saturated buffers, denitrification walls, two-stage ditches, blind inlets, and related structures can substantially reduce nutrient export when they reconnect tile water with reactive soil or engineered treatment zones.
	Comparative effectiveness, cost, and watershed-scale integration	• Compare effectiveness and cost — Wetlands, bioreactors, buffers, drainage management, and other practices all provide meaningful nitrate reductions but differ substantially in cost, phosphorus benefits, habitat value, yield co-benefits, and site requirements. • Scale treatment to watershed targets — Because local treatment benefits attenuate downstream and tile flow can bypass conventional surface best-management practices, large watershed targets require spatial targeting, stacking of source and treatment practices, and monitoring capable of resolving high-flow export.
Other Water-Quality Dimensions: Pesticides, Microbes, Trace Elements, and	–	• Section framing — Pesticides, microbes, trace elements, greenhouse gases, and ecological effects largely follow the same hydrological and preferential-flow controls established for nutrients, with management often exerting stronger control than drainage presence alone.
	Herbicides, Pesticides, and Emerging Contaminants	• Establish pesticide timing and magnitude — Pesticide losses through tiles are usually a very small fraction of application but peak sharply during the first drainage events after application or during seasonal snowmelt.

Greenhouse Gases		• Explain preferential pesticide transport — Macropores explain the rapid appearance of pesticides despite low cumulative mass loss, while transient concentration peaks, degradates, application practices, bioreactor treatment, and emerging contaminants extend this same transport framework.
	Manure-Borne Microbial Contaminants and Resistance Genes	• Characterize microbial and ARG transport — Tile drainage can rapidly transport manure-derived microorganisms and antibiotic-resistance genes, with application method, tillage, soil wetness, and preferential connectivity governing delivery and engineered treatment providing partial attenuation.
	Trace Elements, Selenium, and Acid Sulphate Hazards	• Define trace-element hazard contexts — Drainage can mobilize or concentrate selenium, arsenic, uranium, boron, acidity, aluminium, and other elements in vulnerable settings, making drainage disposal and redox management critical components of system design.
	Greenhouse-Gas Dimensions	• Assess greenhouse-gas coupling — Drainage water can transport substantial dissolved greenhouse gases and influences soil methane and N2O production through aeration and nitrogen cycling, although controlled drainage and buffers can limit some emissions.
	Ecological Responses to Subsurface Drainage	• Assess ecological responses — Drainage modifies soil fauna, birds, wetlands, amphibians, and aquatic habitats in context-dependent ways, with associated land-management practices often producing clearer ecological effects than drainage itself.
Agronomic Benefits: Yield, Salinity Reclamation, and Socio-Economics	–	• Section framing — A large international evidence base shows that drainage increases yield, reclaims saline and waterlogged soils, enables reuse of marginal water, and produces substantial economic and social benefits.
	Crop yield and agronomic outcomes	• Quantify drainage yield gains — Direct comparisons consistently show that drainage raises crop yields by alleviating excess-water stress, with the largest relative gains occurring in severely waterlogged or saline environments. • Evaluate managed-drainage yield benefits — Controlled drainage, subirrigation, and drainage-water recycling can outperform free drainage by retaining or supplying water during dry periods and thereby increasing yield stability. • Explain yield mechanisms and boundaries — Drainage improves root-zone aeration, crop establishment, physiological performance, and water-use efficiency, but benefits depend strongly on soil, crop, weather, water-table depth, and adequate system maintenance.
	Salt leaching and reclamation of saline and waterlogged soils	• Establish reclamation efficacy — Subsurface drainage reliably lowers water tables and removes salts from the root zone, producing large topsoil salinity reductions and corresponding yield improvements across many saline environments. • Define salinity-design controls — Closer spacing and greater drain depth accelerate salt removal up to practical limits, while envelope design and climate-specific geometry strongly influence hydraulic performance and economic optimums. • Explain salt-balance dynamics — Drain effluent is initially salt-rich and becomes fresher as stored salts are depleted, with long-term reclamation governed by leaching fraction, depth, preferential flow, drainage efficiency, and recurring dry-period resalinization.

	Saline and brackish water reuse	• Establish conditions for saline-water reuse — Saline and brackish drainage water can support high crop yields when adequate drainage and leaching prevent root-zone salt accumulation, although sodicity, infrastructure, and disposal constraints remain important.
	Socio-economic returns and investment viability	• Quantify socio-economic benefits — Drainage investments increase crop yield, cropping intensity, household and farm income, employment, land productivity, and in some cases long-term soil quality. • Evaluate financial viability — Formal project appraisals generally show favorable benefit–cost ratios and returns, while larger-scale implementation, optimized layouts, plastic pipe, mechanization, and improved envelopes can substantially reduce installation costs. • Integrate policy and externalities — Because drainage generates downstream water-quality and disposal costs, economically efficient systems require appropriate policy incentives and must increasingly treat drainage-water quality as a design criterion alongside agricultural production.
Modeling, Measurement, and Mapping of Drainage Systems	–	• Section framing — The preceding physical and management conclusions are supported by a mature combination of process modeling, analytical hydraulics, calibration, uncertainty analysis, data-driven methods, geophysics, and remote sensing.
	Field-scale drainage and water-quality models	• Evaluate DRAINMOD performance and limits — DRAINMOD is extensively validated across climates and generally reproduces water tables and drainage well, especially at monthly and annual scales, while daily peaks, site heterogeneity, seepage, and evapotranspiration representation remain important weaknesses. • Compare field-scale model families — Richards-equation models and numerous alternative drainage codes can also simulate water tables and drainage successfully, with no single model universally superior across applications. • Evaluate water-quality model performance — Calibrated drainage and crop models generally reproduce annual nitrate losses and management responses reasonably well but perform less reliably for daily concentrations, residual soil nitrogen, phosphorus, and freeze–thaw dynamics.
	Flow physics: saturated and unsaturated flow and preferential-flow representation	• Establish required flow physics — Purely saturated drainage formulations overpredict discharge because unsaturated storage and vertical resistance matter, while explicit or equivalent representations of macropores are needed to reproduce rapid event responses.
	Analytical drain-spacing solutions and pipe-scale hydraulics	• Refine analytical drain-spacing theory — Transient, evapotranspiration-aware, unsaturated, multilayer, and heterogeneous analytical solutions substantially refine conventional spacing design and often permit more economical layouts than simple steady-state formulations. • Quantify pipe-scale hydraulics — Pipe perforation geometry, envelopes, effective radius, entrance losses, and spatially varied conduit flow materially affect drainage capacity and therefore need explicit hydraulic consideration.
	Watershed representation, sensitivity, calibration,	• Assess watershed drainage representation — Watershed simulations are highly sensitive to how tile networks and subsurface heterogeneity are conceptualized, although simplified

	uncertainty, and evaluation	equivalent or seepage-node representations can reproduce aggregate flow efficiently when appropriately configured. • Identify sensitivity, calibration, and evaluation controls — Hydraulic conductivity and evapotranspiration dominate model sensitivity, while calibration strategy, objective functions, performance metrics, and flow-measurement protocols strongly influence parameter identifiability and conclusions about model skill.
	Data-driven, remote-sensing, and geostatistical methods	• Position data-driven and geospatial methods — Machine learning and statistical surrogates can equal or outperform process models for specific prediction tasks, remote sensing now maps drained extent at regional scales, and geostatistics improves representation of spatially variable design properties.
	Mapping drainage infrastructure and quantifying extent	• Compare field-scale mapping methods — UAV RGB, multispectral, thermal imagery, ground-penetrating radar, and electrical methods can locate buried drains, but effectiveness depends strongly on moisture, soil properties, survey timing, and drain characteristics, making multimethod surveys most reliable. • Scale mapping and design regionally — Machine learning, satellite observations, GIS, groundwater information, and DEM-based tools extend drainage identification and design from fields to watersheds and regions, providing the spatial infrastructure needed for regional assessment.
Conclusions	–	• Synthesize the hydrologic foundation — The evidence converges on drainage as a master hydrological control that redistributes water, raises low flows, shortens travel times, homogenizes catchment response, and thereby establishes the physical basis of subsequent water-quality effects. • Synthesize nutrient-export mechanisms — Nitrate export is dominated by flow-driven subsurface transport while phosphorus export reflects rapid preferential movement of particulate and dissolved forms, linking both major nutrient problems directly back to drainage hydrology. • Synthesize the management hierarchy — Effective mitigation requires a hierarchy of source, drainage-design, and edge-of-field measures whose effects can involve pollution swapping and scale attenuation, making complementary practice stacking necessary for watershed-scale targets. • Synthesize benefits and methodological capability — Environmental costs coexist with well-established yield, reclamation, reuse, and economic benefits, while mature process models, analytical methods, remote sensing, and geophysics provide the tools needed to evaluate these trade-offs. • State the final unifying implication — Convergence across methods, climates, and scales supports the central conclusion that tile drainage is fundamentally a hydrological intervention, so water-quality protection should be treated as a drainage-design objective and managed through coordinated practices across the field-to-watershed continuum.

sssection	subsection	Paragraph role and one-sentence summary
Abstract	–	• Whole-review frontier synthesis — The frontier review argues that tile-drainage research is limited less by a shortage of ideas than by short, spatially narrow, and inconsistent evidence, making long-term harmonised monitoring, broader model validation, network mapping, integrated management, and interdisciplinary adoption research the central agenda.
Introduction	–	• Establish the frontier problem — Tile drainage is essential to agricultural production but also accelerates contaminant delivery, and the review is organized around the field's repeated recognition that major uncertainties remain despite a large literature. • Diagnose the cross-cutting evidence deficit — Across the literature, short records, single sites, limited geographic coverage, and incomplete process representation jointly constrain generalization, so coordinated long-term data and better models must advance together. • Define scope and purpose — The paper complements the state-of-knowledge review by organizing unresolved questions, testable hypotheses, and needed methods and datasets rather than attempting to settle those gaps. • Explain the paper-level progression — The review proceeds from monitoring, modeling, mapping, and scaling foundations through nutrient, water-table, treatment, and salinity frontiers to unintended consequences and integrated socio-economic solutions. • State the unifying research agenda — Progress will require coupling mechanistic understanding with spatial targeting, economics, policy, decision support, and especially longer and more representative datasets rather than relying on any single practice or model.
The Observational Foundation: Longer, Denser, and Harmonised Monitoring	–	• Frame monitoring as a research frontier — Because most drainage conclusions rest on short, site-specific records, the fundamental frontier is building observations that are long, dense, spatially replicated, and harmonized enough to distinguish management effects from climatic and soil variability.
	Extending the temporal record	• Establish the need for long-term records — Long-term monitoring is needed to distinguish treatment effects from climatic noise and slow system change, test whether benefits persist, and evaluate drainage interventions over meaningful multi-year horizons. • Define temporal coverage requirements — Nutrient and hydrologic studies need year-round, high-frequency, multi-year records spanning wet, dry, and normal conditions to resolve seasonal losses, legacy effects, slow soil change, and delayed management responses.
	Broadening spatial and geographic scope	• Test geographic transferability — Site-specific findings must be replicated across contrasting soils, climates, crops, management systems, and regions with complete site characterization before broadly transferable guidance or meta-analysis is possible.
	Monitoring infrastructure,	• Strengthen monitoring infrastructure — Reliable long-term evidence requires robust automated sampling, calibrated sensors, maintenance protocols, water-budget measurements, open datasets, and continually tested data and drainage-network tools。

	instrumentation, and data tools	
	Tracers, isotopes, and source apportionment	• Resolve pathways with tracers and isotopes — Higher-frequency isotope, tracer, source-apportionment, and hydrograph-separation studies are needed to distinguish groundwater, quick-flow, tile, and surface pathways and quantify nutrient sources and water ages.
	Standardisation, harmonisation, and comparability	• Harmonize methods and reporting — Future synthesis depends on standardized monitoring, indicators, site characterization, analytical methods, reporting conventions, shared terminology, and uncertainty assessment that make studies directly comparable.
	Experimental design for detectable effects	• Design studies for detectable effects — Monitoring programs must pair adequate duration with designs capable of detecting treatment effects through paired basins, nested scales, representative replication, controlled experiments, and defensible analysis of unreplicated drainage trials.
Advancing Process-Based Drainage and Water-Quality Models	–	• Frame model credibility as the frontier — Process-based drainage and water-quality models are essential for prediction beyond field trials, but their credibility is constrained primarily by narrow validation domains, missing process measurements, and weak or incomplete model components.
	Broadening the empirical basis for validation	• Broaden validation conditions — Drainage models require independent testing across more soils, climates, management systems, time periods, and design conditions, with assumptions re-evaluated whenever models are transferred to new settings. • Test parameter transferability — A major frontier is replacing site-specific empirical coefficients with functions of measurable site attributes and determining how much local data are needed for reliable model transfer across regions and long design horizons.
	Field and process data still needed to constrain models	• Constrain nitrogen processes with field data — Model improvement requires multi-year measurements of denitrification, soil-N fluxes, carbon–nitrogen dynamics, management lags, redox, and complete water and nutrient budgets to constrain poorly observed nitrogen processes. • Constrain phosphorus and sampling processes — Phosphorus models need better measurements of preferential transport, P sources and forms, high-flow events, soil-P metrics, and sufficiently dense temporal and spatial sampling to capture flashy export and support calibration.
	Preferential and macropore flow representation	• Improve preferential-flow representation — Models need mechanistic, dynamically calibrated representations of macropore connectivity, matrix–macropore partitioning, cracking, shrink–swell behavior, and preferential delivery of water, phosphorus, and solutes across soils and management conditions.
	Coupled hydraulics, seepage, and soil heterogeneity	• Couple drain, ditch, and pore-water hydraulics — A structural modeling gap is the weak coupling of soil-water flow with pressurized or free-surface flow inside drains and ditches, including entrance resistance, backfill transport, network geometry, and multidimensional hydraulics. • Represent heterogeneity, seepage, and groundwater exchange — Residual model error is strongly tied to spatially variable hydraulic properties and poorly closed water balances, motivating depth-resolved conductivity measurements, anisotropic and dual-

		domain representations, seepage dynamics, groundwater data, and improved inverse calibration.
	Nutrient routines, cold-season, and groundwater coupling	• Advance phosphorus routines and tile nutrient pathways — Water-quality models require physically based phosphorus routines that represent macropore delivery, particulate P, fertilizer partitioning, and explicit tile pathways while also improving nitrogen transport phases and complete nutrient budgets. • Improve N cycling, cold-season processes, and groundwater coupling — Priority model gaps include nitrification, denitrification, mineralization, plant N uptake, winter freeze–thaw and snow processes, and coupling drainage with groundwater and reactive transport, especially in cold and arid systems.
	Calibration, sensitivity analysis, and uncertainty	• Strengthen calibration and uncertainty analysis — Model validity requires multiobjective and automated calibration, global sensitivity analysis, parameter-identifiability assessment, independent evaluation, and explicit propagation of input, measurement, parameter, and structural uncertainty.
	Model structure, coupling, and machine learning	• Develop coupled and hybrid model structures — Future models should test coupled hydrologic–biogeochemical systems, data assimilation, machine-learning surrogates, and refined analytical or numerical formulations while validating added complexity across diverse and extreme conditions.
	Instrumentation and measurement-method validation	• Validate measurement methods supporting models — Accurate model evaluation depends on field-calibrated flow structures, sensors, survey methods, and conduit hydraulics, so measurement technologies and hydraulic-loss relationships require systematic validation under realistic conditions.
From Field to Watershed: Mapping Networks and Scaling Response	Overview	• Frame the field-to-watershed spatial problem — The spatial frontier combines two linked problems: mapping largely undocumented tile networks and translating field-scale process understanding and practice performance to watershed scales where water-quality targets are assessed.
	Locating, mapping, and cataloguing drainage networks	• Build drainage-network maps and databases — Improved prediction requires verified tile locations, discharge observations, standardized reporting of depth, spacing, diameter, and mains, and shared drainage-network databases maintained across agencies, researchers, landowners, and contractors. • Advance remote sensing and machine learning for drain detection — Remote sensing research must establish when UAV, multispectral, thermal, SAR, deep learning, and combined sensing approaches reliably detect drains, supported by ground truth, multitemporal imagery, richer covariates, and public training datasets. • Define geophysical applicability — Geophysical mapping needs multi-site evaluation of GPR, EMI, drone, and magnetic configurations so practitioners can predict which methods will work under particular soils, residue, tillage, moisture, and antenna conditions.
	Scaling from field to watershed	• Bridge field processes to watershed response — A central open problem is scaling field-level mechanisms and management effectiveness to watersheds, with intensively monitored small catchments proposed as empirical bridges for model validation and upscaling.

		• Resolve watershed-scale hydrologic effects — Watershed studies must quantify how tiling changes recessions, event response, depressional storage, routing, and surface–subsurface partitioning across scales and test whether representative catchment behavior transfers to other landscapes.
	Channel-network, connectivity, and drain-to-stream processes	• Track drain-to-stream fate — The fate of tile effluent after discharge to ditches and streams remains poorly understood, including its effects on downstream flow, stream power, geomorphology, and drain–channel interactions. • Improve channel-network and sediment process models — Watershed models need explicit reach connectivity, travel times, in-stream biogeochemistry, hillslope–riparian exchange, and non-field sediment sources to represent how tile-derived water and nutrients are transformed downstream. • Quantify connectivity and depressional controls — Management-relevant connectivity metrics, ditch networks, lag times, closed depressions, blind inlets, and antecedent-moisture controls must be incorporated into coupled representations linking individual drains to watershed response.
Managing Nutrient Losses: Nitrogen, Phosphorus, and Cropping Systems	–	• Frame nutrient management optimization — The nutrient frontier is not simply identifying helpful practices but developing the metrics, mechanisms, timing targets, and decision rules needed to optimize nitrogen, phosphorus, and cropping strategies for specific soils, climates, and production systems.
	Refining nitrogen management and the timing of loss	• Refine nitrogen rate, form, timing, and placement — Nitrogen research should optimize fertilizer rate, form, timing, placement, manure crediting, precision management, and 4R implementation to improve nitrogen-use efficiency while balancing crop yield and drainage losses. • Develop next-generation N decision tools — Future N-rate tools should predict economic optima without environmentally excessive application by integrating weather, management history, soil N supply, remote sensing, regional benchmarks, crop sensing, and economic risk. • Resolve application-method questions — Open questions remain around split versus single applications, fall manure, nitrification inhibitors, ammonium stabilizers, fertigation, and simultaneous accounting of all major nitrogen-loss pathways under changing climate and livestock systems. • Credit residual and legacy nitrogen — Management recommendations need better quantification and crediting of mineralizable soil N, residual nitrate, prior legumes, manure, and long-lived legacy N, including their hydrologic and biogeochemical controls after drought or land-use change. • Target the temporal window of nitrate loss — Because nitrate export is concentrated in early spring and other high-loss windows, management and outlet control should be tested specifically during those periods rather than evaluated only through annual averages.
	Phosphorus metrics, mechanisms, and transport	• Define the phosphorus research gap — Subsurface phosphorus remains less understood than nitrogen, requiring better source partitioning, pedologic controls, timing, particulate-size characterization, and simultaneous measurement of surface and tile pathways.

		• Move beyond agronomic soil-test P — Environmental phosphorus risk assessment should incorporate degree of P saturation, sorption capacity, mineralogy, bioavailable P, and isotope-exchange behavior rather than relying on agronomic soil-test P alone. • Resolve desorption, colloids, and particulate transport — Research must quantify how hydrologic fluxes control P sorption and desorption, colloidal and particulate transport, particle mobilization, and bioavailability across alternating wetting, drying, and preferential-flow conditions. • Assess management, event, and legacy-P effects — Phosphorus management needs evidence on tillage, manure timing, climate interactions, small-event losses, source contributions, legacy-P drawdown, multi-species monitoring, and practices capable of reducing nitrate and DRP together. • Resolve phosphorus dynamics in organic soils — Cultivated organic soils require mechanistic studies of phosphorus hotspots, hot moments, chemical forms, microbial turnover, drainage management, and nutrient leaching from abandoned peatlands.
	Cover crops, perennials, and rotational diversification	• Advance cover-crop performance and modeling — Cover-crop research must broaden geographic testing, improve cold-climate establishment and cultivar selection, resolve nitrogen and phosphorus mechanisms and tradeoffs, extend long-term trials, and strengthen winter-rye growth and nutrient algorithms. • Evaluate perennials and diversified rotations — Perennials, bioenergy crops, wetlands, and diversified rotations need stronger causal water-quality evidence, large-scale economic assessment, rotation-scale accounting, and joint evaluation with drainage and nutrient-management practices.
Water-Table Management and Drainage-System Design	–	• Frame active water-table management — The design frontier is a shift from passive drainage toward active, site-specific control of water tables and the irrigation–drainage balance, supported by long-term evidence for yield, water-quality, and operational tradeoffs.
	Controlled drainage, subirrigation, and drainage water management	• Test controlled drainage and active management strategies — Controlled drainage and irrigation should be evaluated as integrated, seasonally adaptive systems, with long-term studies quantifying yield benefits, nutrient effects, costs, management burden, and performance during extreme wet and dry periods. • Coordinate irrigation, seepage, and whole-field water budgets — Effective drainage water management requires coordinated irrigation, site selection, regional water planning, seepage characterization, and complete within- and across-field water and nutrient budgets including recharge and preferential flow.
	Automation, forecast-based, and data-driven control	• Develop automated adaptive control — The next generation of drainage management requires sensor-based control structures, decision-support systems, dynamic water-balance modeling, data assimilation, and control algorithms that jointly consider water quantity, quality, crop growth, and economics. • Integrate forecasts with control operation — Forecast-based drainage management should combine weather uncertainty, evapotranspiration, seepage, subirrigation, local crop conditions, and optimized ditch or outlet schedules into adaptive anticipatory control systems.

	Novel structures, design criteria, and drainage intensity	• Test novel structures and design tools — New drainage structures, multi-level systems, envelopes, perforations, saline-soil layouts, analytical solutions, and CAD/GIS tools require field validation and multi-criteria assessment of hydraulic performance, cost, and site conditions. • Re-evaluate conventional design criteria — Existing drainage depths, spacings, and design standards should be re-tested in situ against crop protection, salinity, drainage volume, controlled traffic, climate change, irrigation modernization, and cost-effectiveness. • Optimize drainage intensity at system scale — Drainage intensity should be optimized as a system-level balance among network capacity, crop and soil response, nutrient loss, soil-parameter uncertainty, and long-term climatic variability rather than as a fixed field specification.
Edge-of-Field Treatment Technologies	Overview	• Frame edge-of-field design frontiers — Edge-of-field technologies have abundant performance demonstrations but still lack transferable rules for sizing, hydraulics, microbial enhancement, reactive media, aging, coupled configurations, and standardized long-term design across climates and landscapes.
	Sizing, scaling, and standardised design	• Scale bioreactor design from site to watershed — Bioreactor research must reconcile site-level efficiency with total watershed load removal through flexible sizing, standardized design procedures, improved loading relationships, hydraulic testing, and validation across full-scale systems beyond a few regions.
	Hydraulics, retention time, and tracer methods	• Improve HRT and tracer-based hydraulic design — Hydraulic residence time remains the key but imperfectly characterized design variable, requiring better flow–removal relationships, transparent porosity estimates, non-Darcy characterization, and reliable conservative-tracer methods.
	Enhancing the microbial engine	• Test bioaugmentation and microbial enhancement — Bioreactor research must determine whether inoculation or microbial engineering can improve cold-temperature nitrate removal, how introduced organisms persist, and whether environmental design is more effective than direct microbial manipulation. • Understand resident microbial controls — A mechanistic microbiome framework is needed to link substrate, HRT, oxygen, nitrate load, pH, community composition, secondary pathways, and by-product formation to treatment performance.
	Reactive media, carbon sources, and structural materials	• Evaluate alternative carbon media and dosing — Alternative carbon substrates, woodchip pretreatment, DOC release, startup flushing, synthetic-carbon dosing, and depth-dependent carbon availability need field-scale evaluation for removal efficiency, longevity, clogging, and low-temperature performance. • Validate envelopes and structural materials — Rice-husk, aggregate, geotextile, filter-sock, and alternative concrete materials require site-specific long-term testing for degradation, acidification, hydraulic behavior, clogging, and structural durability.
	Phosphorus-removal media and filter design	• Optimize phosphorus-removal media — Phosphorus-removal structures need systematic optimization of media chemistry, selectivity, flow rate, contact time, regeneration, sediment control, replacement intervals, cost, and integration with woodchip bioreactors.

		• Resolve dynamic phosphorus behavior in treatment media — Research should determine how redox cycling, polyphosphate-accumulating organisms, startup flushes, aging, dissolved organic phosphorus, and high-frequency DRP dynamics control when treatment materials act as P sinks or sources.
	Aging, longevity, clogging, subsidence, and construction	• Quantify treatment longevity and physical aging — Long-term studies are needed to characterize woodchip decay, biofilm and silt accumulation, carbon depletion, clogging, subsidence, construction effects, replacement thresholds, and network-scale diagnostic methods over multi-decadal service life.
	Pumped, energy-assisted, and coupled configurations	• Test pumped and energy-assisted treatment — Pumped, solar-powered, dual-chamber, bioelectrochemical, and coupled treatment configurations need validation of pumping schedules, source-water targeting, energy costs, hydraulic loading, nutrient fractions, and applicability to new source waters.
	Wetlands, saturated buffers, and companion structures	• Improve saturated-buffer design — Saturated riparian buffers need predictive transport models, landscape replication, vegetation and harvest studies, optimized dimensions, and paired-field economic evaluation to convert demonstrated nitrate removal into reliable design guidance. • Define wetland performance limits — Constructed wetlands require dynamic and long-term studies of residence time, carbon limitation, vegetation management, winter performance, phosphorus maturation, seasonal sizing, and climate-specific treatment efficiency. • Refine companion and combined structures — Bioreactor combinations, controlled drainage, flow regulators, sediment-control structures, hyporheic zones, and other companion practices require field verification of hydraulics, clogging, residence time, and long-term nutrient attenuation across settings.
Salinity, Reclamation, and Drainage-Water Reuse in Arid and Irrigated Systems	–	• Frame the salinity frontier — Salinity research is constrained by slow reclamation trajectories, incomplete observation of salt export, unresolved drainage-water disposal and reuse, and specialized acid-sulfate chemistry that together require improved sensing, modeling, and design.
	Salt-leaching dynamics and long-term reclamation trajectories	• Establish long-term reclamation trajectories — Salt leaching and controlled-drainage effects must be followed over much longer periods to quantify true reclamation rates, salinity–yield tradeoffs, optimal leaching, crop rotations, and cost-effective management under saline irrigation. • Measure salt export and mobilization processes — Salinity programs need continuous drainage and recipient-water monitoring plus process studies of macropore leaching, low-permeability media, replacement between drains, rainfall–irrigation interactions, ponding, and waterlogging–salinity coupling.
	Drainage-water reuse, disposal, and treatment	• Develop safe disposal and treatment pathways — Drainage-water management requires scalable disposal, evaporation, contaminant treatment, trace-element control, real-water testing, life-cycle costing, and near-zero-discharge technologies because soil reclamation simply transfers salts to effluent. • Optimize drainage-water reuse — Reuse research must define crop tolerances, shallow-water-table scheduling, blending, basin-scale salt balances, sequential reuse, treatment

		combinations, and the yield stability and adoption costs needed for economic feasibility.
	Acid sulfate soils and irrigation-return management	• Resolve acid-sulfate and amendment processes — Acid-sulfate systems require better models and maps of sulfide oxidation, hydrologic and nitrogen controls, peat-cover protection, acidity sources, and long-term testing of lime, sulfur, gypsum, biochar, and organic conditioners. • Integrate reclamation with soil quality and irrigation-return management — Reclaimed saline and waterlogged systems need long-term evaluation of soil health, carbon, nutrient and GHG effects, elemental toxicities, root constraints, amendments, crop genetics, return-flow control, and uncertainty-aware irrigation–drainage strategies.
	Sensing, modelling, and design under salinity	• Advance salinity sensing and coupled models — EMI, multispectral and hyperspectral sensing should be integrated with coupled water–salt–nitrate models, root growth, groundwater contributions, climate scenarios, optimization, and large real-world datasets for transferable salinity guidance. • Design adaptive drainage under heterogeneity — Salinity design must account for spatial heterogeneity, preferential drainage economics, pipe configurations, crop protection, drainage-water volume and salinity, and ultimately combine remote sensing with zonal water–salt models for adaptive early warning.
Unintended Consequences and Multi-Objective Tradeoffs	–	• Frame multi-objective consequences — Drainage and treatment can shift pollution among pathways, create by-products, and alter soils over long timescales, yet few studies measure multiple consequences together, leaving the net environmental balance largely unresolved.
	Greenhouse-gas coupling and carbon export	• Expand greenhouse-gas accounting — Drainage studies need coordinated long-term measurements of N_2O, CH_4, carbon sequestration, dissolved gases, wetland carbon storage, and pre-implementation baselines so water-quality benefits can be evaluated against full greenhouse-gas effects. • Quantify indirect N_2O from nitrate loss — Research must resolve nitrate–N_2O linkages and dissolved-gas fate across drainage, groundwater, rivers, and estuaries, develop appropriate emission factors and models, and integrate year-round complete nitrogen and life-cycle balances. • Resolve DOC and treatment carbon budgets — Dissolved organic matter quantity, quality, ecological availability, episodic export, and treatment-structure carbon and methane dynamics require multi-year monitoring and management strategies that account for carbon loss alongside nutrient removal.
	Pollution swapping and by-product formation in treatment systems	• Quantify pollution swapping in treatment systems — Bioreactors and other reductive treatments need high-frequency multi-constituent monitoring of nitrite, mercury methylation, phosphorus release, DOC, and other by-products across startup, aging, redox cycling, and changing operating conditions.
	Emerging contaminants, pesticides, pathogens, and antibiotic resistance	• Broaden assessment to emerging contaminants — Drainage research should extend beyond nutrients to pesticide persistence, seed-treatment exposure, microbial controls, resistance genes, silicon and stoichiometric changes, and other emerging constituents whose sources and ecological effects remain poorly constrained.

		• Model contaminant entry and preferential transport — Studies must quantify how contaminants enter through surface inlets and macropores and improve pesticide models with colloidal, particle-mediated, kinetic, and macropore–drain pathways validated across soils, seasons, and tracer methods. • Resolve pathogen and resistance-gene dynamics — Open questions include pathogen removal mechanisms and residence-time effects, microbial survival in wetlands, manure-derived pathogen persistence, fecal-source tracking, wastewater risks, and the hosts, persistence, and transfer mechanisms of antibiotic-resistance genes.
	Trace-element legacies and ecotoxicology	• Monitor long-lived trace-element and ecological effects — Selenium and other trace-element legacies require decades-long biological and groundwater monitoring, while pesticide ecotoxicology needs mechanistic and field studies of chronic effects on non-target organisms.
	Drainage-induced soil evolution, health, and biology	• Develop soil-health and biological diagnostics — Long-term drainage research should standardize soil biological and health indicators and resolve how hydrology, tillage, compaction, disease, evapotranspiration, crop traits, and management interact to shape soil biological function. • Quantify drainage-driven pedological evolution — Drainage should be treated as a long-term driver of soil evolution, with studies needed to determine the rate, spatial pattern, climate sensitivity, reversibility, and eventual stabilization of mineral, structural, and biological changes.
Toward Integrated, Targeted, and Adoptable Solutions	–	• Frame integration and adoption — Scientific advances will produce outcomes only if practices are combined and targeted effectively, remain resilient under climate change, and are economically, institutionally, and socially adoptable through integrated decision-support systems.
	Bundling, sequencing, and spatial targeting of practices	• Optimize practice bundles — Research should identify synergistic sequences of in-field and edge-of-field BMPs, quantify interactions and legacy effects, and develop targeted combinations capable of meeting water-quality goals rather than searching for a single best practice. • Target hydrologically connected hotspots — Spatial targeting should identify active areas, critical source zones, nitrate-leaching hotspots, and connected pathways, then translate those priorities into GIS-based optimization and basin-scale cost–benefit tools.
	Climate change, extremes, and system resilience	• Design for climate extremes and resilience — Drainage and cropping systems need multi-site, long-term field and modeling tests under drought, flooding, warming, changing precipitation, and multiple climate scenarios to identify designs and practice combinations that remain resilient.
	Economics, incentives, and cost-effectiveness	• Quantify economics and incentives — Future work must compare short- and long-term costs, benefits, opportunity costs, adoption barriers, cost-share mechanisms, nutrient trading, and integrated biophysical–economic performance across drainage and conservation options.
	Policy, institutions, adoption, and the human dimension	• Strengthen governance and participatory management — Effective implementation requires flexible location-specific policy, farmer and stakeholder participation, institutional coordination, education, financing, and governance structures that connect scientific knowledge with local management capacity.

		• Resolve adoption and policy design — Research should test commercial-farm transferability, farmer willingness, participatory experimentation, ecological–economic policy models, pilot programs, legislative barriers, and incentive packages that determine whether conservation practices are actually adopted.
	Integrative landscape-scale frameworks and decision support	• Build landscape-scale decision frameworks — The long-range goal is spatially explicit landscape and watershed models that couple ditch and groundwater processes, ecosystem services, connectivity, practice placement, stakeholder values, and dynamic calibration to optimize multiple outcomes. • Integrate sensing, indices, and shared decision tools — Remote sensing, machine learning, precision agriculture, climate information, improved nutrient and vulnerability indices, trading tools, ecosystem-service models, and richer geo-referenced datasets should be integrated into field-specific, climate-resilient decision support.
Conclusions	–	• Synthesize the evidence constraint — The frontier is defined primarily by insufficient evidence, tools, and coordination to convert promising ideas into transferable guidance, making the observational foundation a prerequisite for progress across all drainage research domains. • Priority 1: coordinated long-term data — The first priority is long, year-round, geographically diverse, harmonized, and openly reported monitoring—supported by tracers, source apportionment, and detectable experimental designs—to separate management effects from climatic and spatial variability. • Priority 2: validated models and spatial scaling — The second priority is broader validation and uncertainty-aware improvement of weak model processes together with drainage-network mapping and field-to-watershed scaling that can support prediction beyond individual sites. • Priority 3: bundled and targeted climate-resilient management — The third priority is to optimize combinations of nutrient, water-table, design, and treatment practices through site-specific targeting, active control, standardized design, and explicit adaptation to drought, flood, and future climate. • Priority 4: interdisciplinary tradeoffs and adoption — The fourth priority is integrated measurement of greenhouse gases, pollution swapping, emerging contaminants, and soil evolution alongside economics, policy, institutions, and farmer engagement so technically effective solutions become viable in practice. • Integrate the four research strands — Long-term data, validated models, targeted multi-objective management, and human-dimension research are mutually dependent, so future progress requires coordinated field campaigns, shared standards and infrastructure, and interdisciplinary integration from mechanism to adoption.